

Regionalization of hydrological modeling for predicting streamflow in ungauged catchments: A comprehensive review

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Abstract

Runoff prediction in ungauged and scarcely gauged catchments is a key research field in surface water hydrology. There have been numerous studies before and since the launch of the predictions in ungauged basins (PUB) initiative by the International Association of Hydrological Sciences in 2003. This study critically reviews and assesses the decadal progress in the regionalization of hydrological modeling, which is the major tool for PUB, from 2000 to 2019. This paper found that the journal publications have noticeably increased in terms of PUB in the past 7 years, and research countries have been expanded dramatically since 2013. The regionalization methods are grouped into three categories including similarity-based, regression-based, and hydrological signature-based. There are more detailed researches focusing on the interdisciplinary and profound improvement of each regionalization method. Namely, tremendous efforts have been made and lots of improvements have been carried out in the parameterization domain for the post-PUB period. However, there is still plenty of room to improve the prediction capability in data-sparse regions (e.g., further verification and proof of multi-modeling adaptation and uncertainties description). This paper also discusses possible research directions in the future, including PUB in a changing environment and better utilization of multi-source remote-sensing information.

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KEYWORDS

hydrological model, parameterization, regionalization, runoff prediction, ungauged basin

1 | INTRODUCTION

Runoff refers to the water flow coming from excess water from rainfall, ice melt, groundwater, and other sources over the earth's land surface. It is the third-largest hydrological flux after precipitation and evaporation in the global land hydrological cycle (Trenberth, Smith, Qian, Dai, & Fasullo, 2007). Runoff is the major water resource for agriculture, industrial, and urban water use, and therefore it has direct impacts on human lives. Because of this, accurate estimation

of surface runoff is not only crucial for understanding complex relationships among various hydrological processes, but it is also crucial for dam/reservoir design as well as water resource planning and management.

The accurate estimates or predictions of catchment runoff mainly rely on hydrological models that play important roles in a wide range of practical issues in society, including water resource management, flood prediction, drought warnings, and coupled systems modeling such as water quality, hydro-ecology, and climate (Randrianasolo, Ramos, & Andréassian, 2011). Successful hydrological modeling often requires accurate inputs such as precipitation and streamflow data. However, high-precision hydrology and meteorological stations cannot be deployed in remote locations. Furthermore, there are very limited hydrological and meteorological observing stations in many developing countries that provide sufficient information for hydrological monitoring and forecasting due to financial limitations. Furthermore, components of the hydrological cycles such as evaporation, infiltration, and subsurface flow are difficult to measure (Sivapalan, 2003). This means that runoff simulations and predictions in ungauged basins (PUB) are one of the most important and challenging issues hydrologists face.

To tackle the PUB issues, scientists often use the hydrological regionalization methods for runoff simulations and predictions in ungauged catchment areas (Zhang & Chiew, 2009a). The idea behind regionalization is to use hydrological model parameters calibrated in the gauged catchment(s) (called donor(s)) to derive the model parameters in a target ungauged catchment, and then predicts runoff time series in the ungauged catchment. Until now, numerous studies for regionalization that have been performed show some apparent progress. However, there are still problems such as unsatisfactory simulation results and low prediction accuracy. This mainly attributes to unclear hydrological similarity indicators, the tremendous heterogeneity of the land surface condition, and the space–time variability of climatic inputs (Sivapalan, 2003).

The most important event regarding PUB was when the International Association of Hydrological Sciences launched the PUB initiative in July 2003 (Sivapalan et al., 2003). The initiative stimulated scientists to put tremendous efforts into ungauged catchments studies, especially on how to parameterize hydrological models in ungauged catchments. This decadal initiative successfully ended in 2012. Following it, several exhaustive reviews were conducted at national (S. X. Liu, Mo, Liu, Xia, & Zhao, 2014; Spence et al., 2013) and international (Hrachowitz et al., 2013; Parajka, Andréassian, et al., 2013; Razavi & Coulibaly, 2013) scales. It has now been more than 7 years since the PUB initiative ended. Thus, it is worth exploring the new prospects and progress that have been made runoff simulation and prediction in ungauged catchments from 2013 onward.

The main aim of this paper is to comprehensively summarize hydrological model regionalization methods and to evaluate related studies for runoff predictions in ungauged catchments, particularly for the post-PUB period (2013 onwards). Furthermore, this paper also aims to discuss challenges and future directions in this domain. The paper is structured as follows: first, three groups of state-of-the-art hydrological model parameterization methods are introduced. Second, the recent progress in model parameterization is summarized by comparing journal papers published in between 2000–2013 and 2013–2019. Third, the applications of these methods for the post-PUB period are summarized and compared. Last, the main challenges and prospects are discussed.

2 | DECADAL RESEARCH PROGRESS IN PUB AREA

This section aims to analyze and summarize the output and outcome progress of predications in ungauged catchment domains in the last two decades, especially for the post-PUB period. Peer-reviewed journal publications were used to display “output” and to predict improvement that is quantified using the mean Nash–Sutcliffe efficiency (NSE) of the leave-one-out cross-validation results over the studied catchments to display “outcome.” Figure 1 compares the number of peer-reviewed journal publications between the two periods of 2000–2012 and 2013–2019 for different countries. The number of papers published in the last 7 years is comparable to that published in 2000–2012 for most of the selected countries. In several countries such as the United States, China, Brazil, and India, the journal publications in the last 7 years have noticeably increased, compared to those in 2000–2012. More importantly, research countries have been expanded dramatically since 2013. Lots of countries have achieved zero breakthroughs, such as Russia, some South American countries other than Brazil, some African countries, and New Zealand (Figure 1). This demonstrates that the development of regionalization is expanding to a larger area and the output is commendable through the PUB initiative. Figure 2 displays annual changes in the total number of publications published in six main hydrological and water resources journals from 2000 to 2019. The figure displays an increasing trend in the number of publications. Comparisons of publications in the six journals show that the Journal of Hydrology published most papers for the hydrological

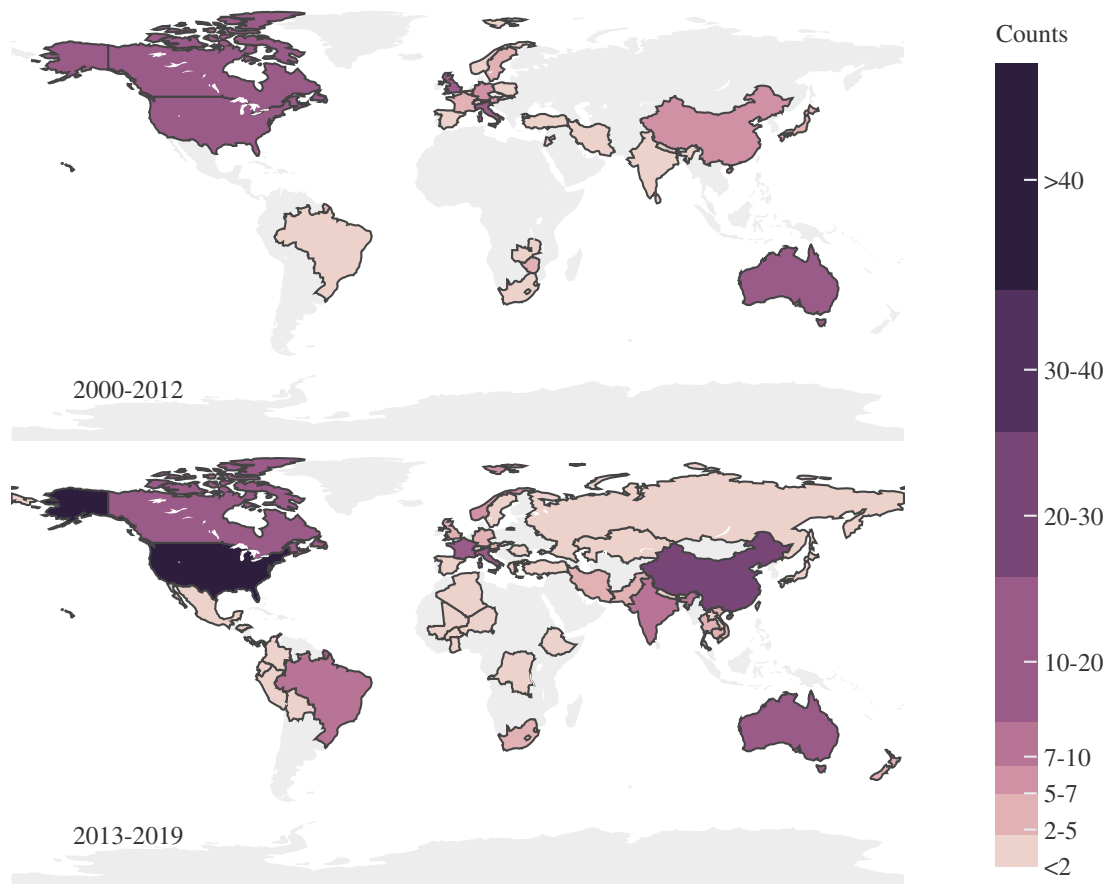


FIGURE 1 Number of peer-reviewed journal publications for the periods of 2000–2012 and 2013–2019. Noted that all selected papers are searched from the ISI Web of Knowledge and Google Scholar with the search term of “hydrological” and “prediction in ungauged basins.” All the papers are then stratified at the country level based on their study region(s)

parametrization, followed by Water Resources Research. It is also noted that the publication number in the *Hydrological Science Journal* has increased since 2013. Figure 3 further compares the outcome of predicted results between the two periods. The predicted results in 2013–2019 are noticeably better than those in 2000–2012, for Australia, Canada, and the United States, but the predicted results are similar in the two periods for China and Europe. A caveat for the NSE comparison is that the various studies used different numbers of catchments and their research periods are not necessarily the same. Therefore, it is inevitable to have some bias in the NSE comparison. This caveat is also applies to Figures 5–7. Nevertheless, it provides a realistic and practical way to compare the outcomes from the numerous studies. In summary, it is somehow unexpected that the prediction in ungauged studies for the post-PUB period have not been weakened, but strengthened instead. Large countries, such as the United States, China, Brazil, and India have put in more effort since 2013 and for lots of countries, where there were no journal publications for the period between 2000 and 2012, they had noticeable publications between 2013 and 2019.

3 | HYDROLOGICAL REGIONALIZATION METHODS

Section 2 has shown that studies on predictions in ungauged catchments are increasing and expanding to further areas around the globe. As the most important process in PUB, hydrological regionalization has experienced a series of evolutions and changes (He, Bárdossy, & Zehe, 2011), and it can be defined as “a collection of methods for transforming hydrological information from gauged areas to ungauged areas” (Oudin, Kay, Andréassian, & Perrin, 2010). The core of parameter regionalization is to “lend” the effective hydrological information from gauged catchments to the ungauged catchments. Furthermore, it not only deals with the transformation of hydrological model parameters but also

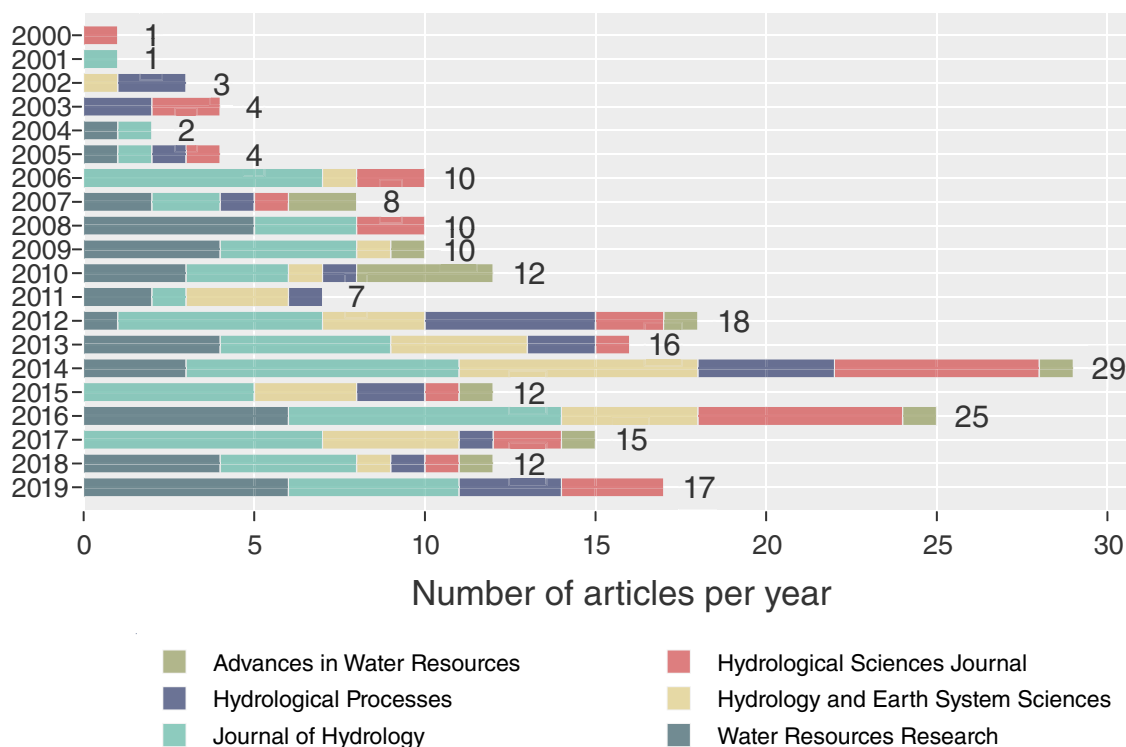


FIGURE 2 Annual change in the number of journal publications for hydrological regionalization

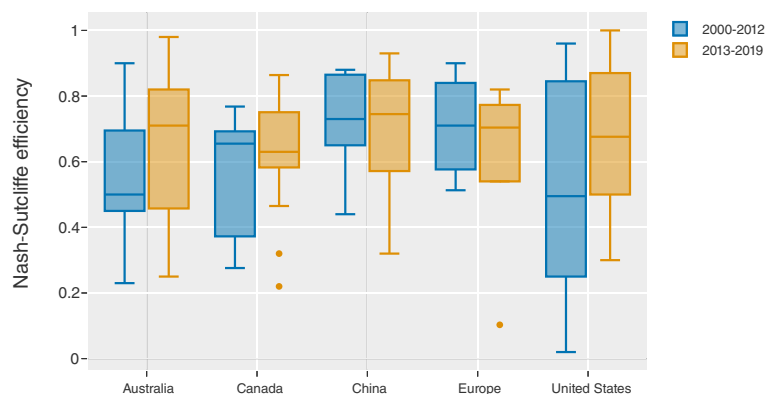


FIGURE 3 Comparison of runoff prediction results between 2000–2012 and 2013–2019 for several countries or continent, where lots of efforts have been put for studying hydrological regionalization in the last two decades (the Nash–Sutcliffe efficiency refers to the mean results of the leave-one-out cross-validation over the studied catchments)

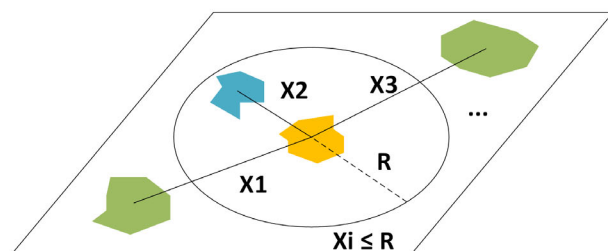
hydrological signatures such as flood frequency distribution, dry season runoff frequency distribution, and runoff duration curve.

There are many classifications of hydrological regionalization. In general, the regionalization method includes three categories: similarity-based, regression-based, and hydrological signatures-based. The similarity-based regionalization implements parameter regionalization through different catchment similarity measures (Merz & Blöschl, 2004). For example, the spatial proximity method is proposed based on catchment spatial similarity that adjacent spaces have autocorrelation characteristics (Anselin, 2013). The physical similarity method is put forward based on the assumption that similar climatic and land cover conditions have similar hydrological characteristics. Unlike the former, the regression-based regionalization does not need to define the similarity measure. Instead, it establishes the relationship between hydrological parameters and catchment characteristics. Some hydrologists transform hydrological signatures from gauged catchments to ungauged catchments and inversely calculate the hydrological model parameters using the Monte Carlo method to achieve predictions in ungauged catchments. The details of the above methods are shown in Figure 4, with their basic principles and concepts. In the last two decades, many combinations and comparative studies of different regionalization have been conducted (Kay, Jones, Crooks, Calver, & Reynard, 2006; H. Li & Zhang, 2017; Merz & Blöschl, 2004; Parajka, Blöschl, & Merz, 2007; Swain & Patra, 2017b; Yang, Magnusson, Rizzi, & Xu, 2018;

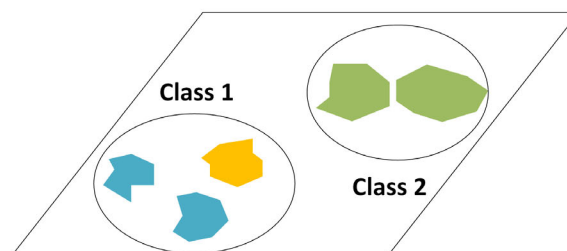
Hydrological Parameterization Methods

Similarity- based Regionalization

◆ Spatial Proximity



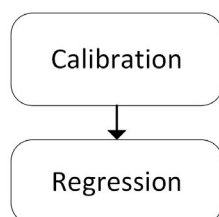
◆ Physical Similarity



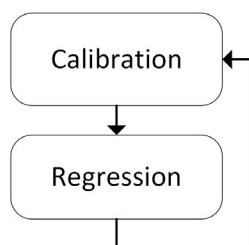
■ Donor Catchment ■ Target Catchment ■ Non-Donor Catchment

Regression-based Regionalization

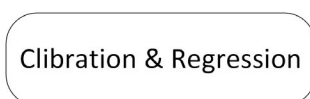
◆ Two-step Regression



◆ Sequential Regression

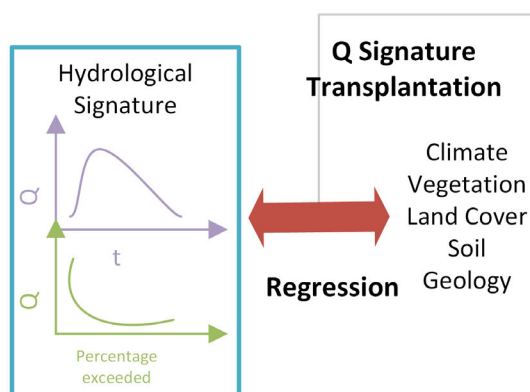


◆ One-step/simultaneous Regression



Hydrological Signature based Regionalization

Donor Catchment



Target Catchment

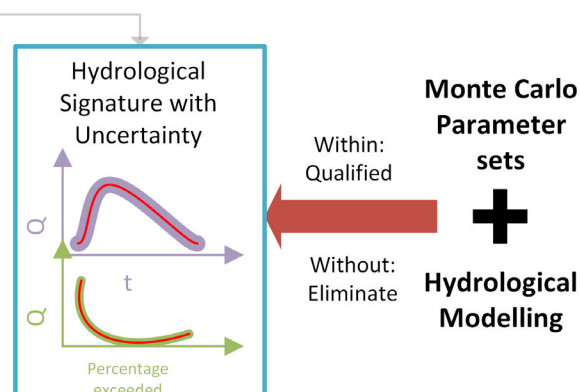


FIGURE 4 The concept map of hydrological regionalization methods for runoff prediction in ungauged catchments (the concept based on the regression method refers to He et al., 2011; the main idea of the hydrological signature-based method comes from Yadav, Wagener, & Gupta, 2007)

Yang, Magnusson, & Xu, 2018; Young, 2006; Zhang & Chiew, 2009a, 2009b). However, it is difficult to determine the best regionalization method because there exists large uncertainty when using different hydrological models and input data. In the following sections, the pros and cons as well as the developmental prospects in the post-PUB period of each regionalization approach are comprehensively summarized.

3.1 | Similarity-based methods

3.1.1 | Spatial proximity

The spatial proximity refers to finding one or more donor catchments (gauged) that are adjacent to the target catchment (ungauged) in space, and regionalizing its parameters to the target catchment by interpolation or averaging. The spatial proximity has two regionalization methods (i.e., simple proximity and spatial interpolation). The simple proximity often uses Euclidean distance to calculate the distance between the donor catchment and the target catchment, and the hydrological model parameters of the target catchment area provided by the nearest one or several donor catchments. Vandewiele and Elias (1995) have used this method to parameterize the model in 75 river catchments in Belgium and found that 44% of the river catchments achieved good results. Randrianasolo et al. (2011) discovered that in France, neighbor catchments can contribute to providing forecasts of good quality ungauged sites, and that it required at least five neighbor donors to gain good model simulation results. On the other hand, spatial interpolation is more complicated than simple proximity. In general, the spatial interpolations are Kriging (Stein, 2012) and inverse distance weighting (Shepard, 1968), which are widely used in parameter regionalization application and research (Merz & Blöschl, 2004; Parajka, Merz, & Blöschl, 2005; Samuel, Coulibaly, & Metcalfe, 2011; Ssegane, Tollner, Mohamoud, Rasmussen, & Dowd, 2012a). In some areas, spatial interpolation has more acceptable results for regionalization (i.e., less degradation from calibration to regionalization) than other approaches. For example, Merz and Blöschl (2004) have mentioned that spatial proximity in Austria is more effective than the regression-based method. However, spatial proximity is based on the assumption that adjacent catchments have similar underlying surfaces and meteorological conditions (Blöschl, 2006). This means it does not apply to areas where these conditions change drastically and complexly in space (G. Ali, Tetzlaff, Soulsby, McDonnell, & Capell, 2012; Reichl, Western, McIntyre, & Chiew, 2009). There is another limitation when using spatial proximity, which is that the density of gauged streamflow stations may affect the accuracy of regionalization (H. E. Beck et al., 2016). In other words, spatial proximity may not obtain satisfactory regionalization results in the areas with sparse gauges. McIntyre, Lee, Wheeler, Young, and Wagener (2005) found that when using only one nearest neighbor donor catchment for transposition, the regionalization results were not satisfactory. He also discovered that a catchments spatial proximity cannot fully represent the consistency of hydrological characteristics of the target catchment (Shu & Burn, 2003).

In the post-PUB period, spatial proximity techniques are still widely used and more comparisons between spatial proximity and other regionalization methods have been implemented around the world. Swain and Patra (2017a) assessed four regionalization techniques on 32 catchments in India and the results obtained from the spatial proximity were found to be superior to the regression-based methods. Yang, Magnusson, Rizzi, et al. (2018) made a comprehensive assessment of the regionalization methods in the high latitudes of Norway and found that distance-based similarity approaches (spatial proximity, physical similarity) performed better than regression-based approaches. Arsenault, Breton-Dufour, Poulin, Dallaire, and Romero-Lopez (2019) also evaluated three regionalization methods in Mexico and discovered that the donor-based spatial-proximity regionalization method was to be preferred in semi-arid and humid ungauged catchments. These results agree with previous conclusions. However, Pugliese, Farmer, Castellarin, Archfield, and Vogel (2016) compared spatial proximity and regression-based methods for predicting flow duration curves (FDCs) in the southeastern United States and revealed that the two methods perform similarly throughout flow-regimes, which differs to previous studies. Recently, some researches (de Lavenne, Boudhraa, & Cudennec, 2015; de Lavenne, Skoien, Cudennec, Curie, & Moatar, 2016) compared spatial proximity and geomorphology-based approach that has the advantage of being more physically based. Their final results demonstrated similar performances with both approaches, showing that the geomorphology-based inversion approach is as reliable as the spatial proximity approach. Different from the regionalization comparison studies, the studies on the spatial proximity approach combining with other regionalization methods like physical similarity (Razavi & Coulibaly, 2017), regression-based (Castellarin et al., 2018; Steinschneider, Yang, & Brown, 2015) method, as well as new techniques such as data assimilation (Pugliese et al., 2018), machine learning (Hong, Zhang, Wang, Qian, & Hu, 2017), have also gradually appeared in the

last decade. At the same time, new improvements based on traditional spatial proximity approaches such as three-dimensional canonical Kriging (Castellarin, 2014) and the streamflow–streamflow (Q–Q) method (Andréassian, Lerat, Le Moine, & Perrin, 2012) showed good performance in some areas, which needs further evaluation.

3.1.2 | Physical similarity

The physical similarity method involves catchment classification. It refers to a regionalization approach for finding one or more donor catchments that have similar catchment attributes (such as climate type, land use, and geological conditions) to the target catchment. This method is also based on the physical distance between the donor catchment and target catchment as well as based on the assumption that the natural geographic and climatic conditions can determine the hydrological characteristics of the catchment (Acreman & Sinclair, 1986; Burn & Boorman, 1993; Merz, Blöschl, & Parajka, 2006; Oudin et al., 2010). In the physical similarity method, the similarity index for finding the donor catchments or clustering analysis are two ways to generate a set of model parameters that are to be applied to the target catchment. Some frequently used similarity index definitions are shown in Table 1. The proper definition of a similarity index and its criterion is the key basis for this type of method. In addition to the similarity indexes in Table 1, another class of similarity index that is been widely used is the topographic index (K. J. Beven & Kirkby, 1979) and soil-topographic index (K. Beven, 1986). It is assumed that points with the same topographic index value have the same hydrological response to rainfall. Based on this, it is proposed that catchments with a similar topographic index (or soil-topographic index) frequency distribution are hydrologically similar (Kong & Rui, 2003). Yao, Zhang, and Li (2013) have applied the Xin'anjiang model to the Tunxi catchment and pointed out that the rainfall-runoff model parameters can be transferred to when the terrain index distribution curve is similar. It is also found that the topographic index can excavate catchments' similarity (L. Liu, 2006). It is worth noting that topographic indices have not yet been applied to large-scale regionalization, and their applicability needs to be further tested. In addition, similar measures based on copula have also been proposed (Samaniego, Bárdossy, & Kumar, 2010). Some studies have applied it in regional analysis and it is proved to be very promising because of its possibility of describing the full stochastic dependence or relationship between catchments (Chowdhary & Singh, 2010). Another physical similarity is based on a clustering analysis to divide the catchments into different groups. After clustering or classifying catchments, there are two ways of regionalization with different applicable conditions. The first way is to uniformly calibrate all catchments in the same clustering group and directly transpose the most similar donor catchment parameters to the target catchment (Parajka et al., 2005; Zhang & Chiew, 2009a), which is suitable for areas with less spatial heterogeneity. The second way is to select the donor catchments within the threshold and transpose its parameters to the target catchment by weighted average (Kay et al., 2007), which can be used in areas with more complex land cover.

The systematic clustering analysis based regionalization is getting more attention compared to the similarity index in the post-PUB decade. This is because more researchers have realized that hydrological similarity is essential when implementing regionalization in different areas and it is not easy to categorize catchments without a series of physical analysis framework. Agarwal et al. (2016) proposed a multiscale entropy method using a wavelet transform and k-means based hybrid approach for the clustering of hydrologic catchments. Papageorgaki and Nalbantis (2016) first

TABLE 1 Several common catchment attribute similarity indicators

Sources	Equation
Burn and Boorman (1993)	$Dist_{t,d} = \sum_{i=1}^n \frac{ CD_{di} - CD_{ti} }{\Delta CD_i}$
Kay, Jones, Crooks, Kjeldsen, and Fung (2007)	$Dist_{t,d} = \sqrt{\sum_{i=1}^n \left(\frac{CD_{di} - CD_{ti}}{\sigma_i} \right)^2}$
He et al. (2011)	$Dist_{t,d} = \sqrt{\sum_{i=1}^n \omega_i \left(\frac{CD_{di} - CD_{ti}}{\sigma_i} \right)^2}$

Note: Among them, the CD is catchment attributes, d is donor catchment, t is target catchment, n is the number of catchment attributes, ΔCD_i is the i th catchment attribute range, σ_i is the standard deviation of the i th catchment attribute, ω_i is the weight coefficient of the i th catchment attribute.

performed unsupervised classification through the fuzzy c-means method using hydrological signatures and then performed supervised classification through the k-nearest neighbor method. S. K. Singh, McMillan, Bardossy, and Fateh (2016) explored a new measure of similarity using a data depth function and then clustering catchments using the affinity propagation clustering algorithm. Kanishka and Eldho (2017) used the Isomap nonlinear dimensionality reduction technique along with the nonlinear principal component analysis, principal component analysis, and k-means cluster analysis to improve the efficiency of watershed classification. Apart from more comprehensive classification analysis, researches on physical similarity regionalization are not only aiming at regionalizing hydrological model parameters, but also more targets such as flow regimes (Solans & Mellado-Diaz, 2015), observed streamflow (Agarwal et al., 2016; Choubin et al., 2019; Yilmaz & Onoz, 2020), and hydrological signatures (Chouaib, Alila, & Caldwell, 2019; Elesbon et al., 2015; Papageorgaki & Nalbantis, 2016; Westerberg et al., 2016).

3.1.3 | Donor catchment selection

For similarity-based regionalization methods, the choice of the number of donor catchments will directly affect the simulation and prediction results of the target catchment. Therefore, the relationships between the donor catchment number and model evaluation criterion such as the NSE coefficient should be analyzed. The number of donor catchments may be related to research region, station density (Yang, Magnusson, Rizzi, et al., 2018), the hydrological model (Oudin, Andréassian, Perrin, Michel, & Le Moine, 2008), and the regionalization method (spatial proximity or physical similarity). It is suggested that 5–6 donor catchments are required for accurate model predictions (H. Li & Zhang, 2017; Oudin et al., 2008; Swain & Patra, 2017b). In addition to regionalization, usage of multiple donor catchments can effectively reduce random errors in comparison to the usage of a single donor catchment (Arsenault, Poissant, & Brissette, 2015; Randrianasolo et al., 2011; Zhang & Chiew, 2009a).

3.2 | Regression-based methods

Regression-based regionalization can be traced back to 1960 when Nash (1960) tried to find the correlation between the unit hydrograph and the catchment attributes. At the same time, Dalrymple (1960) had proposed the idea for flood frequency curve transposition. The main core of regression-based regionalization is to establish a certain regression relationship between model parameters and catchment descriptors. This relationship is used to predict the model parameters of target catchments, and then to achieve the purpose of a simulation or prediction runoff in ungauged catchments. He et al. (2011) divided regression-based regionalization into three categories. The first is the two-step regression, which is also the earliest approach of parameter regionalization (H. E. Beck et al., 2016). It first calibrates model parameters for donor catchments, respectively; and then establishes the relationship between parameters and catchment descriptors. The second is sequential regression, which first determined the most recognizable parameter and sequentially calibrated the last parameter. The third is the one-step simultaneous regression, where the calibration and regression are regarded as an integral process with the objective function to be to perform regionalization.

3.2.1 | Two-step regression

The two-step regression is the most intuitive parameter regionalization approach, and there have been many studies so far (Abdulla & Lettenmaier, 1997a, 1997b; Heerdegen & Reich, 1974; Heuvelmans, Muys, & Feyen, 2006; Jarboe & Haan, 1974; Karlinger, Guertin, & Troutman, 1988; Sefton & Howarth, 1998; Servat & Dezetter, 1993; Wagener & Wheeler, 2006; Waylen & Woo, 1984; W. Weeks & Ashkanasy, 1983; W. D. Weeks & Boughton, 1987; Xu, 2003; Yokoo, Kazama, Sawamoto, & Nishimura, 2001). Despite this, this method still has some issues such as the large ignorance of a high correlation between model parameters (Kuczera & Mroczkowski, 1998) and less representative results of linear regression due to multicollinearity in catchment attributes (Blöschl, 2006). In addition, the “equifinality” problem in the model calibration process may directly impact the effect of regression (Kokkonen, Jakeman, Young, & Koivusalo, 2003; McIntyre et al., 2005). Furthermore, the runoff would be poorly predicted in ungauged catchments

when there is no obvious regression relationship between model parameters and the catchment attributes (Fernandez, Vogel, & Sankarasubramanian, 2000; Kim & Kaluarachchi, 2008).

3.2.2 | Sequential regression

The sequential regression is proposed to solve the “equifinality” problem in the hydrological model (He et al., 2011). In the calibration step, the parameters are sorted in a certain order based on the judgment of the hydrological mechanism or the response relationship between the parameters and the objective function (Calver et al., 2005). Only one parameter is calibrated and then regression, in order, in each loop until the last parameter. The idea of this method comes from the multistep automatic calibration scheme (Hogue, Sorooshian, Gupta, Holz, & Braatz, 2000). The advantage of sequential regression is that it can enhance the identifiability of model parameters (He et al., 2011). However, there is also a drawback which is once a parameter is selected, the parameter space will also be reduced. So, this solution is typically not as good as calibrating the entire parameter set directly.

3.2.3 | One-step/simultaneous regression

The one-step regression combines parameter calibration and regression. This method simultaneously calibrates all objective functions in the parameter regionalization process. So, it is also called simultaneous regression. It was first proposed by Fernandez et al. (2000) when he was using the monthly water balance model for the regionalization of parameters in the south-east United States. An overall objective function combining parameter calibration effects and regression equations is proposed as:

$$\text{Maximize} \left(\left[\frac{1}{m} \sum_{i=1}^m R_i^2 \right] + \left[\frac{R_a^2 + R_b^2 + R_c^2 + R_d^2}{4} \right] \right), \quad (1)$$

where m is the number of catchment sites, R_i^2 is the simulation effect of the i th site observations and where the model simulation values, R_a^2 , R_b^2 , R_c^2 , and R_d^2 represent the regression of model parameters and catchment attributes equation fitting effect (here, the “abcd” model has four parameters a , b , c , and d , so there are four effective regression parameters).

However, studies have shown that this one-step regression does not have much improvement in the simulation results compared with the two-step regression. Unless the understanding of the mechanism can be deepened no matter what kind of statistical methods such as single-objective or multiobjective functions are utilized (the difference is very small in the simulation effect).

During the PUB decade, He et al. (2011) randomly selected 15 regression-based regionalization studies to find out the widely used effective catchment attributes, which indicated that the accuracy of the regression-based regionalization method largely depends on the choice of catchment attributes. In the post-PUB decade, there are still some studies focusing on catchment attribute selection, but more emphasis has been put on some indices that can characterize the changing underlying such as land cover change variables (Ochoa-Tocachi, Buytaert, & De Bievre, 2016). Furthermore, some studies have started to evaluate landscape variables such as topography and soil impacts on the regionalization of nonstationarity hydrological responses under climate change (Teutschbein, Grabs, Laudon, Karlsen, & Bishop, 2018; Visessri & McIntyre, 2016). Recently, new regression methods have been proposed and different regression approaches have also been more comprehensively assessed. For example, the Gaussian process regression was assessed and compared with traditional linear regression and artificial neural network models in 438 catchments across the United States, and it was found that it outperformed both regression methods in most cases (Sun, Wang, & Xu, 2014). The newly proposed constrained hydrologic regression technique that compared with multiple linear regression (MLR) and stepwise regression in Pakistan has proven to be a more promising tool in PUB (Waseem, Ajmal, & Kim, 2016). Zhang, Chiew, Li, and Post (2018) first introduced the regression tree ensemble approach and assessed it with MLR, log-transformed MLR, and hydrological modeling in 605 catchments across Australia, which outperformed the two linear regressions in predicting signatures of flow dynamics. Apart from that, there also have been some studies that tried to

combine regression-based regionalization with clustering analysis (Huang, Wang, Zheng, Yu, & Xu, 2015), machine learning (Prieto, Le Vine, Kavetski, García, & Medina, 2019), and spatial proximity approach (Steinschneider et al., 2015).

3.3 | Hydrological signature methods

Hydrological signature(s) refers to static (average streamflow, flow percentile, flood frequency, etc.) and dynamic (FDC slope, baseflow index, flow change rate, etc.) indicators that can reflect the hydrological characteristics of catchments on different time scales (Westerberg & McMillan, 2015; Zhang et al., 2018). In recent years, the study of the hydrological signature has attracted more and more attention, because it has many applications, such as catchment characterization and classification (Ssegane, Tollner, Mohamoud, Rasmussen, & Dowd, 2012b; Wagener, Sivapalan, Troch, & Woods, 2007), runoff prediction in an ungauged catchment (Shu & Ouarda, 2012; Zhang, Vaze, Chiew, Teng, & Li, 2014), water resources assessment planning and management (Biondi & De Luca, 2017), and ecological and environmental flow assessment (H. J. Beck & Birch, 2012; Newcomer et al., 2012). In addition, H. E. Beck, De Roo, and van Dijk (2015) made a different hydrological signatures global map based on measures of runoff data, which was then used for simulation evaluation and optimization of large-scale hydrological models.

There are two main ways to apply hydrological signatures in regionalization. The first is to directly perform regionalization of hydrological signatures (especially FDCs) from donor catchments to target catchments, which has been more frequently implemented in the post-PUB (Atieh, Taylor, Sattar, & Gharabaghi, 2017; Chouaib et al., 2019; Gyawali, Griffis, Watkins, & Fennessey, 2015; Kult, Fry, Gronewold, & Choi, 2014; Ochoa-Tocachi et al., 2016; Pugliese et al., 2016; Qamar et al., 2016; Viglione et al., 2013; Worland, Farmer, & Kiang, 2018; Zhang et al., 2018). The superiority of regionalizing the hydrological signature is that it can directly characterize the hydrological characteristics in many aspects and scales, but it is particularly important to select the appropriate hydrological signature for specific research (Addor et al., 2018). The study Prieto et al. (2019) conducted in northern Spain shows that only a small part of the 103 hydrological signatures can represent 87% of the hydrological changes in these catchments. This proves that pre-evaluation catchment attributes can effectively improve the ability to characterize hydrological processes when using a hydrological signature. Another issue when using hydrological signatures in regionalization is that different signatures do not necessarily provide independent information and could lead to signatures being omitted or included on a subjective basis. To solve this, Almeida, Le Vine, McIntyre, Wagener, and Buytaert (2016) presented a method that accounted for the inter-signature error correlation structure so that regional information was neither neglected nor double-counted when multiple signatures were included. This showed that the consideration of such structure may improve predictions when error correlations are strong. In addition, some hydrological signatures may have multiple ways to be. For example, H. E. Beck et al. (2015) introduced four calculation methods for the base flow index in his research. This means it is crucial to choose an appropriate way to calculate a single hydrological signature. The second application of hydrological signatures in regionalization is to constrain model parameter sets, then use them to predict streamflow in ungauged catchments. This hydrological signature-based regionalization method has already been implemented during the PUB decade. For example, Yadav et al. (2007) proposed a regionalization scheme based on hydrological signatures (Figure 1). Specifically, they established a relationship between 28 hydrological signatures and 13 catchment attributes in the donor catchments. Then the hydrological signature ranges was further determined in target catchments after hydrological signature transposition based on that relationship and adding uncertainty. Next, the Monte Carlo method was used to randomly generate many parameter sets to run the hydrological model to obtain the time series and the corresponding hydrological signatures were further calculated in target catchment. If it was within the target hydrological signature range, the parameter set was marked as qualified; otherwise, the parameter set was rejected. Finally, all qualified parameter sets would be further evaluated for prediction in ungauged catchments. At the early post-PUB decade, there were some studies which applied this method (Costa, Fernandes, & Naghettini, 2014; Pinheiro & Naghettini, 2013; Shu & Ouarda, 2012; Westerberg et al., 2014). However, this research has been relatively rare in the post-PUB decade since it is more complex and might be limited by the accuracy of meteorological data compared with direct regionalization of hydrological signatures (H. E. Beck et al., 2016). In regionalization, studying the relationship between hydrological signatures and catchment attributes seems to be more direct than when studying the relationship between hydrological model parameters and catchment attributes, which can deepen the understanding of the hydrological cycle mechanism. The research of regionalization based on hydrological signatures should make profound progress soon.

4 | REGIONALIZATION STUDIES COMPARISON

Since lots of effort have been made during the period of 2013–2019. This section only summarizes the progress on comparing regionalization made in this period. To explore the prediction effect and adaptability of different parameter regionalization methods in different regions and evaluate whether the prediction effect is related to the catchment number or aridity index, this review summarizes 33 typical application studies (Table 2) after 2013 (the end of the PUB initiative) using different regionalization methods. The selection of these 33 studies is based on the following considerations:

- whether various regions are included and only 2–3 typical studies are selected for the same region;
- whether different parameter regionalization methods or hydrological models are applied;
- whether there are clear results of parameter regionalization effect (with NSE as the standard since it is the most widely used evaluation criterion).

Figure 5 summarizes the performance of different parameter regionalization methods and in the 33 selected studies, the median calibration of the NSE is 0.75. This is an overall calibration, and it cannot represent any hydrological simulation consequence in a specific area. Among the regionalization methods, the median of the NSE for physical similarity regionalization is the highest (0.69), and the median regionalized of the NSE based on the regression method is the lowest (0.66). It is worth noting that the median of the NSE of spatial proximity and physical similarity is very close, but regionalized NSE based on spatial proximity has a wider range. In addition, the 25th–75th percentile of the NSE range of the physical similarity method is small, while that of the regression-based method is the largest. It can be illustrated from above, that the physical proximity regionalization is relatively more stable than the other two methods. The spatial proximity and regression-based regionalization methods can perform well in some regions, while in others it cannot.

To better evaluate the adaptability of each regionalization method after calibration, Figure 6 compares calibration of the NSE and regionalization of the NSE using a scatterplot. The 1:1 line represents the same NSE of calibration and regionalization. If a point is located closer to the 1:1 line, this point has better adaptability in regionalization. As can be seen in Figure 6, the trend line of regression-based NSE is further away from the 1:1 line compared with that of the spatial proximity of the NSE and physical similarity of the NSE. This indicates that the regression-based method has the lowest adaptability among the three methods. Interestingly, the trend line of spatial proximity of the NSE almost coincides with that of physical similarity of the NSE. It is noted that when the calibration of the NSE is larger than 0.75, the physical similarity method shows good adaptability.

The aridity index, defined as mean annual potential evaporation divided by mean annual precipitation, represents the degree of dryness of the climate at a given location. Figure 7 examines the relationship between the NSE and the aridity index, and it shows that the NSE tends to decrease when aridity increases for all regionalization methods. This is consistent with the findings of Parajka, Viglione, et al. (2013). This indicates that it is harder to predict runoff in dry regions, which is also consistent with the general knowledge of runoff prediction in different climate zones. Compared to other regions, Australia is a dry country, where the lowest regionalization of the NSE is obtained from three approaches, except for one case where the NSE of 0.01 is obtained from the regression approach for Spain (that could be an abnormal value). Furthermore, Figure 7 shows the NSE variation is relatively stable with an increase in aridity index. But, the physical similarity and regression approaches show more noticeable degradation with the increase in aridity index. This suggests that more attention should be paid when choosing these two approaches.

Figure 8 evaluates whether the regionalization effect is related to the catchment number (gauging stations). The size of the bubble represents the number of catchments in Figure 8. It can be seen that the adaptability of the regionalized NSE (i.e., the difference between calibration and regionalization) is related to the number of catchments. The countries that have more gauged catchments such as the United States and Canada are significantly better than other countries such as Ethiopia, Iran, and Korea, where the number of gauged catchments are sparse. This illustrates that the accuracy of regionalization increases with an increasing number of gauged catchments or the runoff site density.

Such comparison and analysis is valuable for parameter regionalization and prediction in ungauged catchments since useful information can be extracted from the synthesis of existing studies. However, it is not straightforward to compare the results of different studies (Parajka, Viglione, et al., 2013). Various studies in different climate and underlying conditions use different input settings, hydrological models, and evaluation criteria, which makes it difficult to consistently assess the regionalization performance of these studies across the whole world. For further evaluation of regionalization in ungauged catchments, the unified regionalization result and conclusion format as well as clear comparative tables or figures would be more helpful.

TABLE 2 33 typical studies for studying parameter regionalization methods and their regionalization results across the globe from 2013 to 2019 (among these studies, if there are multiple hydrological models, the NSE results are listed with comma corresponding with each model)

Study	Region	Catchment number	Hydrological model	Time scale	Model calibration NSE		Parameter regionalization NSE		
					Calibration	Validation	Spatial proximity	Physical similarity	Regression-based
Choubin et al. (2019)	Iran	40	IHACRES, SWAT	Monthly	0.62, 0.65	0.56, 0.64		0.05, 0.61	
Arsenault et al. (2019)	Mexico	30	GR4J, MOHYSE, HMETs	Daily	0.76, 0.8, 0.79		0.69–0.71, 0.69–0.73, 0.5–0.71	0.67–0.7, 0.7–0.72, 0.5–0.71	0.68, 0.7, 0.5
Prieto et al. (2019)	Spain	92	PDM	Daily					–0.65 to 0.68
Zhang et al. (2018)	Australia	605	SIMHYD	Daily					0.06–0.88
Castellarin et al. (2018)	Danube Basin, Europe	511		Daily			0.93		0.57–0.76
Tegegne and Kim (2018)	Ethiopia and Korea	13	SWAT	Daily	0.49, 0.62	0.44, 0.53	0.27, 0.42	0.25, 0.5	
Chang, Kjeldsen, McIntyre, and Lee (2018)	Korea	19	PDM	Daily	0.7	0.58			0.42
Yang, Magnusson, Rizzi, et al. (2018)	Norway	118	WASMOD	Daily	0.86	0.81	0.79	0.8	0.67
H. Li and Zhang (2017)	Australia	605	SIMHYD and Xinanjiang	Daily			0.55, 0.66	0.56, 0.67	
Poissant, Arsenault, and Brissette (2017)	Canada	266	GR4J	Daily	0.7–0.85		0.68–0.75	0.7–0.76	0.6–0.7
Swain and Patra (2017a)	India	32		Daily	0.7	0.63	0.58	0.51	0.48
Hsu and Huang (2017)	Taiwan	22		Annually					0.35–0.98
Razavi and Coulibaly (2016)	Canada	90	MAC-HBV, SAC-SMA	Daily			0.33–0.62, 0.32–0.62		
Arsenault and Brissette (2016a)	Canada	264	HSAMI	Daily	0.81	0.79	0.6–0.7	0.63–0.7	0.6–0.7
Song et al. (2016)	China	13	TVGM	Monthly	0.90	0.91			0.75
Ergen and Kentel (2016)	Turkey	13	Drainage-area ratio (DAR)	Daily			0.53		
Pugliese et al. (2016)	United States	182		Daily			0.57		0.66

TABLE 2 (Continued)

Study	Region	Catchment number	Hydrological model	Time scale	Model calibration NSE		Parameter regionalization NSE		
					Calibration	Validation	Spatial proximity	Physical similarity	Regression-based
Parajka, Merz, Skoien, and Viglione (2015)	Austria	555		Daily			0.7		
Qamar, Ganora, and Claps (2015)	Italy	118		Monthly			0.74	0.81	
Waseem, Ajmal, and Kim (2015)	Pakistan	8					−0.53–0.78		−0.52 to 0.79
Steinschneider et al. (2015)	United States	73	abcd	Monthly, yearly			0.78		0.75
Smith, Marshall, and Sharma (2014)	Australia	118	PDM	Daily	0.72		0.73	0.73	
Zhang et al. (2014)	Australia	228	GR4J	Daily			−0.26–0.81	0.14–0.74	
Arsenault and Brissette (2014)	Canada	268	HSAMI	Daily	0.84		0.69–0.75	0.70–0.75	0.66
F. Li et al. (2014)	China	8	SIMHYD and GR4J	Monthly	0.87, 0.8	0.86, 0.83	0.71, 0.66	0.69, 0.71	
Castellarin (2014)	Italy	41		Daily			0.89		
R. Singh, Archfield, and Wagener (2014)	United States	87		Daily	0.75			0.6–0.8	
Sun et al. (2014)	United States	438		Monthly					0.45–0.5
Archfield, Pugliese, Castellarin, Skoien, and Kiang (2013)	United States	61		Annually			0.8		
Piman and Babel (2013)	Thailand	2	HEC-HMS	Sub-daily	0.952				0.855
Mendicino and Senatore (2013)	Italy	19		Daily					0.6–0.92
Wallner, Haberlandt, and Dietrich (2013)	Germany	41	HBV-IWS	Daily	0.62–0.7	0.44–0.56		0.54–0.56	
Livneh and Lettenmaier (2013)	United States	220	ULM	Daily	0.539	0.523	0.41–0.43		

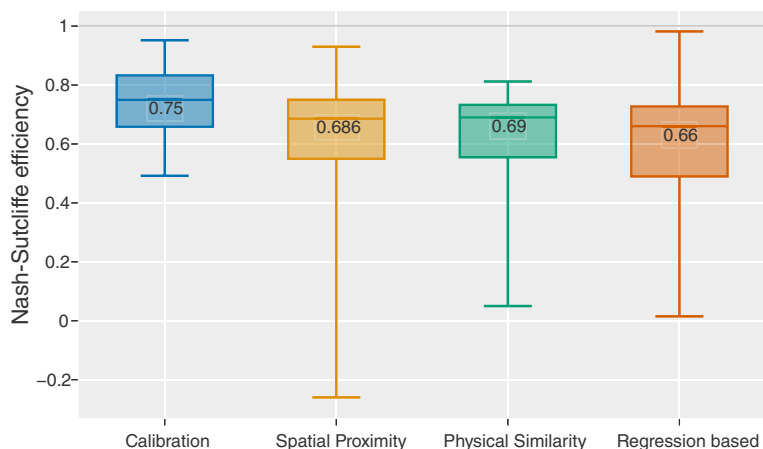


FIGURE 5 Nash–Sutcliffe efficiency (NSE) comparison among different parameter regionalization methods (the NSE of each specific regionalization method refers to the mean of the leave-one-out cross-validation results over the studied catchments)

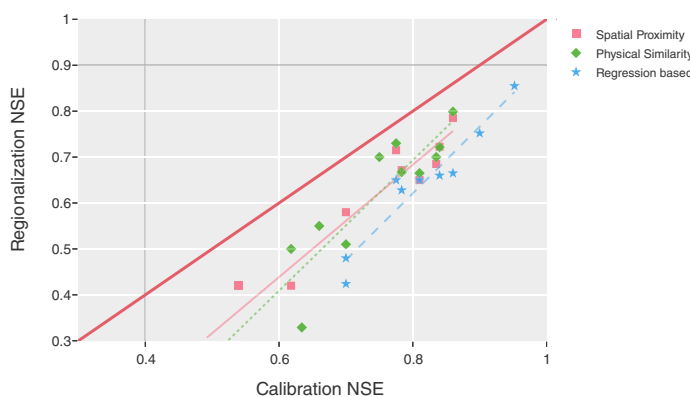


FIGURE 6 Calibration of the Nash–Sutcliffe efficiency (NSE) and regionalization of the NSE scatter plot (Q signature method is not listed due to few research cases; the calibration of the NSE refers to the mean of calibration results over the donor catchments and the regionalization of the NSE refers to mean leave-one-out validation results of the target catchments)

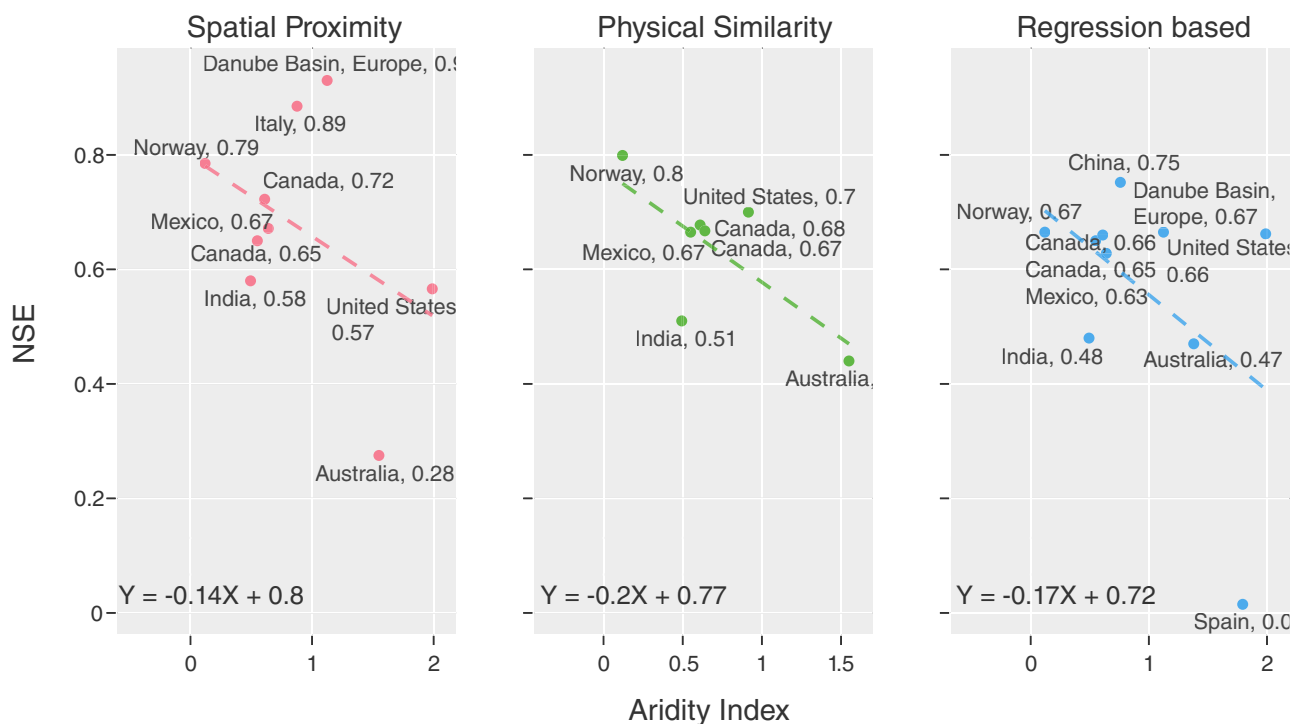
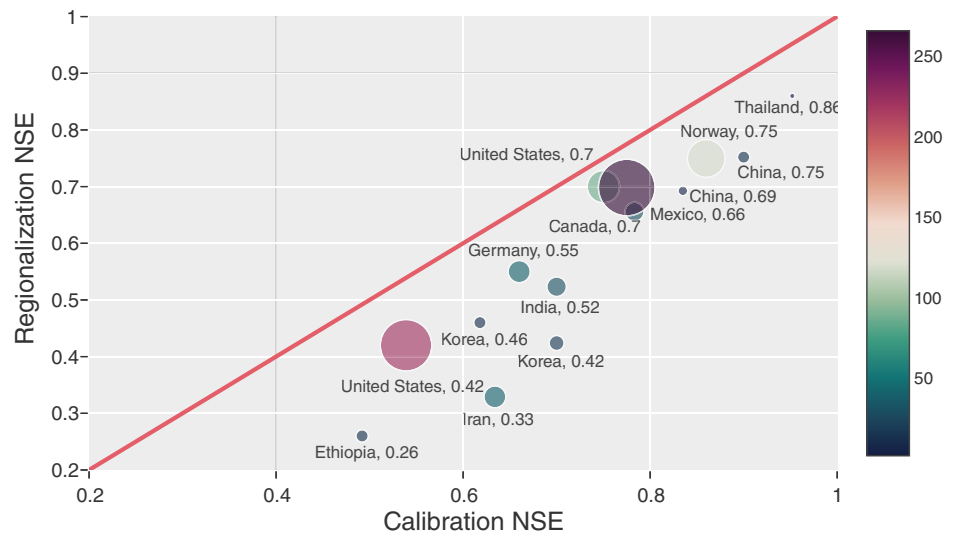


FIGURE 7 Scatter plot of aridity index (PET/P) and the Nash–Sutcliffe efficiency (NSE) regionalized using spatial proximity, physical similarity, and regression-based approach (the regionalization of the NSE refers to mean of leave-one-out validation of the NSE over the target catchments)

FIGURE 8 Parameter regionalization process of the Nash–Sutcliffe efficiency (NSE), calibration period of the NSE and catchment number bubble diagram



5 | CHALLENGES AND FUTURE OUTLOOK

5.1 | Major challenges

Though lots of progress for parameter regionalization have been made in the last two decades, few studies have concluded the structural adaptability analysis of different hydrological models and in different regions. The accuracy of most hydrological simulations in an ungauged catchment based on parameter regionalization depends largely on the model structure. Different kinds of hydrological models and structures should be adapted when facing different climate and underlying conditions. The choice of the hydrological model and structure can be determined based on the prior knowledge of the input data, catchment attributes, and catchment hydrological characteristics. From a macro perspective, there are currently many pieces of research on model adaptability analysis at catchment or regional scale. Especially in the post-PUB period, studies are starting to assess the possibility of using multi-model averaging techniques for better hydrological simulation in ungauged catchments. Razavi and Coulibaly (2016) applied a multi-modeling approach in 90 watersheds across Ontario, Canada, and they found that the combined model was more efficient and appeared more robust compared to the individual models. However, Arsenault and Brissette (2016b) got contradicting results. Using multi-model averaging in Québec, Canada, he discovered that multi-model averaging techniques were not necessarily suited for regionalization applications and that models selected in such studies must be chosen carefully in order to be as robust as possible. Multi-modeling can improve the hydrological simulation in some places, but simple averaging does not guarantee better adaptability than a single best model simulation. Thus, finding a better multi-modeling scheme, or exploring and summarizing the general model structure that can be adapted on the intercontinental scale and even the global scale based on existing research is a huge challenge. From a micro perspective, whether the proposed model structure represents the hydrological cycle characteristics of a target catchment and the characteristics of water balance across the soil-vegetation-atmosphere (SVA) system in a single catchment, is also a part that needs to be refined and further studied.

Generally, the uncertainties of regionalization mainly come from two sources: (a) uncertainty of hydrological modeling or hydrological signature calculations for the gauged catchments, (b) uncertainty stemming from the regionalization methods and procedure. In the first uncertainty source, the “equifinality” problem is defined as having multiple, differing parameter sets that are equally acceptable during the model calibration and validation processes (K. Beven, 2006). However, there are research results that indicate that although set parameter selection can play an important role in regionalization method performance, “equifinality” does not contribute significantly to the overall uncertainty witnessed throughout the regionalization methods applications in the post-PUB decade (Arsenault & Brissette, 2014). This is worth having further discussions about, and the conclusion would be more clear if there were more focused studies on it. In the second aspect, there is different uncertainty origin in different regionalization methods. In spatial proximity, the number of donor catchments will affect the parameter regionalization results in some areas. The determination of the nearest neighbor range will also bring certain uncertainty to the regionalization results.

In physical similarity, the definition of catchment similarities and the inconsistent number of catchment attributes selected in different regions will also affect the final results. In addition, several studies have indicated that there is the uncertainty of regionalization in changing conditions (R. Singh, van Werkhoven, & Wagener, 2014; R. Singh, Wagener, Werkhoven, Mann, & Crane, 2011). Some of them evaluated temporal stability and transferability of hydrological model parameters and suggested that spatiotemporal transfer of hydrological model parameters have the potential to be a viable option for climate change-related hydrological studies (Patil & Stieglitz, 2015). However, further research is still needed to explore the relationship between spatial and temporal aspects of catchment hydrological variability.

When using physical similarity or regression-based regionalization, an implicit assumption that similar catchment attributes should have similar hydrological characteristics is included but it is an intuitive judgment. The two are not strictly equal. Therefore, whether the catchment attribute characteristics can directly represent the hydrological response is the first issue to verify. Moreover, how to apply catchment attributes to characterize hydrological similarity is also a problem that needs to be solved (Choubin et al., 2019). Which combinations of catchment attributes best represent hydrological characteristics, whether unnecessary catchment attributes should be added, and how to describe the correlation between catchment attributes are all core questions that need to be considered when studying the relationship between catchment attributes and hydrological characteristics. The existing catchment similarity indicators are described by attribute-based Euclidean distance. Whether it is possible to find an indicator that can represent the process similarity to better characterize the hydrological similarity is also a challenge in parameter regionalization study.

The quality of hydrological data sources and the density of hydrological sites are factors that directly affect the simulation of large-scale hydrological models and the parameter regionalization process. H. Li and Zhang (2017) have found that the grid parameter regionalization provides huge uncertainty in the central part of Australia because there are rare gauging sites and scarce runoff data. Therefore, a larger density of measured runoff data should be used in order to ensure the best parameter calibration result, which is the most important step in hydrological modeling. As Blöschl (2006) has mentioned in his summary—"perhaps the best way to solve the problem of runoff simulation in the ungauged catchment is to build more hydrological sites." Recently, studies also suggested that the use of a limited number of discharge observations for regionalization can be another way to solve the data scarcity problem (Lerat et al., 2012; Pool, Viviroli, & Seibert, 2019).

5.2 | Future outlook

The hydrological cycle is mainly affected by climate and underlying conditions. The 2014 Intergovernmental Panel on Climate Change report pointed out that the global climate is changing (Pachauri et al., 2014). Besides, human activities such as logging, afforestation, and soil-water conservation measures have also changed the underlying surface conditions of natural catchments. Therefore, whether parameter regionalization in a changing environment is still applicable and how runoff simulations in an ungauged catchment in a changing environment should be improved are worthy to discuss. Some studies have begun to analyze regionalization in a changing environment. For example, Yang, Magnusson, and Xu (2018) found that the prediction result difference derived from various regionalization methods may increase in future scenarios, and that physical similarity regionalization is more stable. Similar research and evaluation work should be developed in the future to advance the process of runoff simulation in an ungauged catchment and large-scale hydrological simulation. In addition, whether the hydrological similarity classification will be affected in a changing environment is also a question worthy of study since the catchment under the changing underlying surface or meteorological conditions may show different hydrological characteristics from the past (such as deforestation or bushfires). Under such changing conditions, the existing catchment classification results may be different at different spatial scales, and whether different classification may affect the runoff prediction in the ungauged catchments should also be further studied.

Reasonable catchment classification can simplify and synthesize the understanding of hydrological characteristics, which is the cornerstone of parameter regionalization. It is the new trend to use advanced statistical analysis tools such as big data and machine learning to find hydrological regional characteristics in large-scale or even global scale. By reasonably defining the similarity matrix or similarity index, the hydrological similarity can be better portrayed, and the catchment classification strategy can be improved. Effective catchment classification at a global scale can summarize the specific hydrological characteristics in each classification, allowing hydrologists to find suitable hydrological models based on different hydrological characteristic types and sharing existing simulation experiences to further collaborate to improve runoff simulation and prediction in an ungauged catchment.

The large-scale parameter regionalization process and streamflow prediction in ungauged catchment needs more comprehensive multi-source information and more powerful computation tools for it to be supported. In model simulation and calibration, multi-source remote sensing data such as vegetation, soil moisture, and ET can add dynamic information to traditional hydrological models, which can greatly improve the hydrological simulation effect in areas and increase the reliability of the calibrated parameters (Alvarez-Garreton et al., 2016; Gui et al., 2019; Silvestro et al., 2015). In addition, as the input of a hydrological model, mutual verification and comparison of station precipitation data and satellite precipitation data can also improve the input accuracy (Gao et al., 2017; Mei, Nikolopoulos, Anagnostou, & Borga, 2016; Xue et al., 2013). In physical similarity and regression-based regionalization, a variety of remote sensing information such as land use and vegetation cover also needs to extract the catchment attributes. The multi-disciplinary cross of the SVA system and the coupling of multi-source data such as soil, vegetation, and precipitation can more comprehensively characterize the complexity and diversity of the water cycle in a catchment. Remote sensing data can also be used directly to calibrate model parameters in ungauged catchments without streamflow observation (Kunnath-Poovakka, Ryu, Renzullo, & George, 2016) or constraining model parameters or hydrological signature (Kunnath-Poovakka, Ryu, Renzullo, & George, 2018). For example, Zhang et al. (2020) proposed a new approach to calibrate the hydrological model using solely remotely sensed evapotranspiration, and the calibrated parameters set, to predict the streamflow time series directly. Since this approach does not use observed streamflow for model calibration, it does not require regionalization. This approach performs very well in wet regions of Australia and has a large potential for data-sparse or ungauged regions (Huang et al., 2020). Apart from calibration using remote sensing data, physically based hydrological numerical models that do not need to calibrations are also developing fast these years due to recent advances in computing power and data availability (Lewis, 2016). This is a new field, providing a new research area for the hydrological community to investigate how to precisely forecast real-time streamflow with big dates and powerful computational ability. It should be a good open question for the hydrological community to discuss and investigate.

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AUTHOR CONTRIBUTIONS

Yuhan Guo: Conceptualization; formal analysis; software; validation; visualization; writing-original draft. **Yongqiang Zhang:** Conceptualization; formal analysis; resources; supervision; writing-review and editing. **Lu Zhang:** Supervision; writing-review and editing. **Zhonggen Wang:** Supervision.

CONFLICT OF INTEREST

The authors have declared no conflicts of interest for this article.

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APPENDIX A.

Table A1 shows 135 studies related to hydrological model regionalization across the globe from 2013 to 2020. The detailed information is for those who are further studying hydrological model regionalization, and expecting to quickly find similar research studies within their interests. Note that the regionalization studies put forward a new regionalization framework (e.g., MPR technique and SPED framework), and are listed in the regionalization methods column. As mentioned before, the target of hydrological model regionalization can be a model parameter, observed streamflow as well as hydrological signature. So, the regionalization target column is essential for understanding the purpose of the regionalization. Apart from the basic regionalization target categories such as model parameters, FDCs, and hydrological signature, we have also included some less frequent regionalization targets that clearly state the paper-like design flood and flood quantiles to ensure consistency with the original text. Papers are mainly focusing on catchment grouping or clustering itself, and they are marked as catchment classification in the regionalization target column.

TABLE A1 135 studies of hydrological model regionalization including the information of the country, catchment (gauging stations) number, time scale, hydrological model, regionalization methods as well as target across the globe for the period of 2013–2020

First author	Country	Stations/ catchments	Time scale	Hydrological model	Regionalization methods	Regionalization target
Archfield et al. (2013)	United States	61	Annual peak- streamflow	—	Spatial proximity	Design flood
Kumar, Livneh, and Samaniego (2013)	United States & Germany	81	Daily and hourly	mHM	Multiscale parameter regionalization (MPR) technique	Model parameter
Livneh and Lettenmaier (2013)	United States	220	Daily	ULM	Regression-based, regional calibration	Model parameter
Mendicino and Senatore (2013)	Italy	19	Daily	—	Regression-based	FDCs
Piman and Babel (2013)	Thailand	7	Daily and hourly	HEC-HMS	Regression-based	Model parameter
Pinheiro and Naghetini (2013)	Brazil	17	Daily	RIO GRANDE	Regression-based	Calibration of model parameter by regionalized FDCs

TABLE A1 (Continued)

First author	Country	Stations/ catchments	Time scale	Hydrological model	Regionalization methods	Regionalization target
Pomeroy, Fang, Shook, and Whitfield (2013)	Canada	1	Daily	CRHM (platform)	Physical reasoning	Model parameter
Pruski, Nunes, Pruski, and Rodriguez (2013)	Brazil	7/1	Daily	HBV-IWS	Physical similarity	Model parameter
Ricard, Bourdillon, Roussel, and Turcotte (2013)	Canada	49/1	Daily	Hydrotel	Regional calibration	Model parameter
Viglione et al. (2013)	Austria	213	Daily	HBV	Spatial proximity	Hydrological signatures
Wallner et al. (2013)	Germany	41	Daily	HBV-IWS	Physical similarity	Model parameter
M. Ali et al. (2014)	United States	50	Daily	—	Regression-based	the storage–discharge relationship
Arsenault and Brissette (2014)	Canada	268	Daily	HSAMI	Spatial proximity, physical similarity, regression-based	Model parameter
Castellarin (2014)	Italy	41	Daily	—	Spatial proximity	FDCs
Cibin, Athira, Sudheer, and Chaubey (2014)	United States	2	Daily	SWAT	Physical similarity	Model parameter
Costa et al. (2014)	Brazil	2	Daily	RIO GRANDE	Extended Burr XII probability distribution based method	Calibration of the model parameter by regionalized FDCs
Drogue and Plasse (2014)	France	149	Daily	GR4J	Four distance-based methods	Daily hydrographs
Kult et al. (2014)	United States, Canada	163	Monthly	LBRM, MESH	Regression-based	Hydrological signatures
Laaha, Skoien, and Blöschl (2014)	Austria	8,000	Daily	—	Spatial proximity, regression-based	Hydrological signatures
F. Li et al. (2014)	China	8	Monthly	SIMHYD and GR4J	Spatial proximity, physical similarity	Model parameter
Palanisamy and Workman (2014)	United States	37/1	Daily	TOPMODEL	Regression-based	Averaged transfer function
Patil and Stieglitz (2014)	United States	756	Daily	EXP-HYDRO	Spatial proximity, physical similarity	Model parameter
Pugliese, Castellarin, and Brath (2014)	Italy	18	Daily	—	Spatial proximity	FDCs
Sellami, La Jeunesse, Benabdallah, Baghdadi, and Vanclooster (2014)	France	10	Daily	SWAT	Physical similarity	Model parameter
R. Singh, Archfield, et al. (2014)	United States	87	Daily	A lumped rainfall-runoff model	Physical similarity	Model parameter
Smith et al. (2014)	Australia	118	Daily	PDM	Spatial proximity, physical similarity	Model parameter
Sun et al. (2014)	United States	438	Monthly	—	Regression-based	Observed streamflow
Wang et al. (2014)	China	42	Monthly	SWBM	Regression-based	Model parameter
Westerberg et al. (2014)	Honduras, Nicaragua, Costa Rica, Panama	36	Daily	WASMOD	Physical similarity	FDCs for constraining model predictive uncertainty
Ye et al. (2014)	United States	50	Daily	—	Regression-based	The storage–discharge relationship

(Continues)

TABLE A1 (Continued)

First author	Country	Stations/ catchments	Time scale	Hydrological model	Regionalization methods	Regionalization target
Zhang et al. (2014)	Australia	228	Daily	GR4J	Spatial proximity, physical similarity	Hydrological signatures
Arsenault et al. (2015)	Canada	267	Daily	HSAMI	Spatial proximity, physical similarity, regression-based	Model parameter
H. E. Beck et al. (2015)	Global	3,000–4,000	Daily, monthly	—	Machine learning	Hydrological signatures
Biondi and De Luca (2015)	Italy	1	Sub-daily	A combination of the SCS-CN and the Nash' model	Regression-based	Hydrological signatures
de Lavenne et al. (2015)	France	6	Yearly, hourly	A geomorphology- based model	Spatial proximity	Hydrological observations
Elesbon et al. (2015)	Brazil	61	Daily	—	Physical similarity	Hydrological signatures
Gyawali et al. (2015)	United States	93	Daily	—	Regression-based	Hydrological signatures
Hailegeorgis, Abdella, Alfredsen, and Kolberg (2015)	Norway	26	Hourly	First-order nonlinear system model, HBV, basic grid model	Spatial proximity, physical similarity	Model parameter
Huang et al. (2015)	China	15/1	Daily	BTOPMC	Regression-based	Model parameter
Muller and Thompson (2015)	Austria, Nepal	109	Daily	—	TopREML	Hydrological signatures
Nruthya and Srinivas (2015)	India	17	Daily	—	Regression-based	FDCs
Parajka et al. (2015)	Austria	555	Daily	—	Spatial proximity	Observed streamflow
Patil and Stieglitz (2015)	United States	294	Daily	EXP-HYDRO	Spatial proximity	Model parameter
Pechlivanidis and Arheimer (2015)	India	42	Daily	India-HYPE	Machine learning	Model parameter
Pruski, Rodriguez, Nunes, Pruski, and Singh (2015)	Brazil	6/1	Annually	—	Regression-based	Hydrological signatures
Qamar et al. (2015)	Italy	118	Monthly	—	Spatial proximity, physical similarity	Hydrological signatures
Skaugen, Peerebom, and Nilsson (2015)	Norway	84	Daily	DDD	Regression-based	Model parameter
Solans and Mellado- Diaz (2015)	Spain	1	Annually	—	Classification	Flow regimes
Steinschneider et al. (2015)	United States	73	Monthly, yearly	abcd	Spatial proximity, regression-based	Model parameter
Viviroli and Seibert (2015)	Switzerland	49	Hourly	PREVAH	Spatial proximity, physical similarity	Model parameter
Wan and Konyha (2015)	United States	25	Daily	LinRes	Spatially averaging	Model parameter
Waseem, Ajmal, et al. (2015)	Pakistan	8	—	—	Spatial proximity, regression-based	FDCs
Waseem, Shin, and Kim (2015)	Korea	14	Annually	—	Spatial proximity	FDCs
Agarwal et al. (2016)	United States	530	Monthly	—	Physical similarity (clustering)	Observed streamflow
Almeida et al. (2016)	United States	84	Daily	PDM	Regression-based	Hydrological signatures
Arsenault and Brissette (2016a)	Canada	264	Daily	HSAMI	Spatial proximity, physical similarity, regression-based	Model parameter

TABLE A1 (Continued)

First author	Country	Stations/ catchments	Time scale	Hydrological model	Regionalization methods	Regionalization target
Arsenault and Brissette (2016b)	Canada	267	Daily	HSAMI, MOHYSE, HMETs	Spatial proximity	Model parameter
Athira, Sudheer, Cibil, and Chaubey (2016)	United States	8	Daily	SWAT	Regression-based	PDF of model parameters
H. E. Beck et al. (2016)	Global	1787	Daily	HBV	Physical similarity	Model parameter
Bock, Hay, McCabe, Markstrom, and Atkinson (2016)	United States	5,457	Monthly	MWBM	Physical similarity	Model parameter
de Lavenne et al. (2016)	France	184	Daily	WFIUH	Spatial proximity	Hydrographs
Ergen and Kentel (2016)	Turkey	13	Daily	DAR method	Spatial proximity	Model parameter
Farmer (2016)	United States	182	Daily	DAR, QPPQ method	Spatial proximity	Observed streamflow
Lebecherel, Andreassian, and Perrin (2016)	France	609	Daily	GR4J	Spatial proximity	Model parameter
Ochoa-Tocachi et al. (2016)	Venezuela, Colombia, Ecuador, Peru, and Bolivia (Tropical Andes)	24	Daily	—	Regression-based	Hydrological signatures
Papageorgaki and Nalbantis (2016)	Greece	24	—	—	Physical similarity (clustering)	Hydrological signatures
Pruski, Rodriguez, Pruski, Nunes, and Rego (2016)	Brazil	7/1	Long-term average	—	Regression-based	Long-term average flow
Pugliese et al. (2016)	United States	182	Daily	—	Spatial proximity, regression-based	FDCs
Qamar et al. (2016)	Italy	124	Daily	—	Physical similarity	FDCs
Razavi and Coulibaly (2016).	Canada	90	Daily	MAC-HBV, SAC- SMA	Spatial proximity	Model parameter
S. K. Singh et al. (2016)	New Zealand	21	Hourly	TopNet	Physical similarity (clustering)	Model parameter
Song et al. (2016)	China	13	Monthly	TVGM	Regression-based	Model parameter
Vergara et al. (2016)	United States	1,672	Hourly	The kinematic wave model	Multidimensional statistical modeling	Model parameter
Visessri and McIntyre (2016)	Thailand	42	Monthly	—	Regression-based	Hydrological signatures
Waseem, Ajmal, and Kim (2016)	Pakistan	8	Annually	—	Regression-based	FDCs
Waseem, Ajmal, Kim, and Kim (2016)	Korea	14	Daily	—	Spatial proximity, physical similarity	Hydrological signatures
Westerberg et al. (2016)	United Kingdom	43	Hourly	—	Physical similarity	Hydrological signatures
Zhou et al. (2016)	China	12	Daily	VIC-3L	Spatial proximity, physical similarity, regression-based	Model parameter
Atieh et al. (2017)	United States	260	Daily	—	Machine learning	FDCs
Barbarossa et al. (2017)	Global	1885	Annually	PCR-GLOBWB	Regression-based	Mean annual flow
Biondi and De Luca (2017)	Italy	8	Daily, sub-daily	A combination of the SCS-CN and the Nash' model	Regression-based	Hydrological signatures

(Continues)

TABLE A1 (Continued)

First author	Country	Stations/ catchments	Time scale	Hydrological model	Regionalization methods	Regionalization target
Clark, Ahn, and Palmer (2017)	United States	14	Daily	HSPF, HBV	Regression-based	Model parameter
Hailegeorgis and Alfredsen (2017a)	Norway	26	Daily	—	Regression-based	Flood quantiles
Hailegeorgis and Alfredsen (2017b)	Norway	26	Hourly	—	Regression-based	FDCs
Hamel, Guswa, Sahl, and Zhang (2017)	Australia, United States	730	Monthly	DWBM	Regression-based	Model parameter
Hong et al. (2017)	China	7	Annually	—	Spatial proximity, machine learning	Observed streamflow
Hsu and Huang (2017)	Taiwan	22	Annually	—	Regression-based	FDCs
Jillo, Demissie, Viglione, Asfaw, and Sivapalan (2017)	Ethiopia	10	Daily	HBV-Light	Regression-based	Model parameter
Kanishka and Eldho (2017)	United States	68	—	—	Physical similarity (clustering using NLPCA and PCA)	Catchment classification
Lachance-Cloutier, Turcotte, and Cyr (2017)	Canada	75	Daily	Hydrotel	Spatial proximity	Observed streamflow
H. Li and Zhang (2017)	Australia	605	Daily	SIMHYD and Xinanjiang	Spatial proximity, physical similarity	Model parameter
Ndzabandzaba and Hughes (2017)	Swaziland, South Africa	122	Monthly	Pitman model	Regression-based	Hydrological signatures
Poissant et al. (2017)	Canada	266	Daily	GR4J	Spatial proximity, physical similarity, regression-based	Model parameter
Razavi and Coulibaly (2017)	Canada	90	Daily	MAC-HBV, SAC-SMA	Spatial proximity, physical similarity, machine learning	Model parameter
Requena, Ouarda, and Chebana (2017)	Canada	151	Daily	—	Regression-based	Flood quantiles
Swain and Patra (2017a)	India	32	Daily	—	Spatial proximity, physical similarity, regression-based	FDCs
Tanase (2017)	Romania	12	Daily	—	Regression-based	FDCs
Araujo, de Mello, Gollin, de Quadros, and Gomes (2018)	Brazil	18	Annually	—	Spatial proximity	Hydrological signatures
Ayzel (2018)	Russia	8	Daily, monthly	HBV, GR4J, SIMHYD	Regional calibration	Model parameter
Burgan and Aksoy (2018)	Turkey	109	Annually	—	Regression-based	FDCs
Castellarin et al. (2018)	Danube Basin, Europe	511	Daily	—	Spatial proximity, regression-based	FDCs
Chang et al. (2018)	Korea	19	Daily	PDM	Regression-based	Model parameter
Chouaib, Alila, and Caldwell (2018)	United States	73	Daily	SAC-SMA	Physical similarity	Model parameter
Farmer and Levin (2018)	United States	66	Daily	—	Spatial proximity	Observed streamflow
Ouarda, Charron, Hudecha, St-Hilaire, and Chebana (2018)	Canada	190	Daily	—	Generalized additive models (GAMs), spatial proximity, regression-based	Low-flow characteristics

TABLE A1 (Continued)

First author	Country	Stations/ catchments	Time scale	Hydrological model	Regionalization methods	Regionalization target
Perez, Mantilla, and Krajewski (2018)	United States	74	Monthly	-	Regression-based	Hydrological signatures
Pugliese et al. (2018)	Tyrol, Europe	46	Daily	E-HYPE	Spatial proximity	FDCs
Qamar et al. (2018)	Italy	124	Daily	—	Spatial proximity, physical similarity	FDCs
Sengupta et al. (2018)	United States	26	Hourly	HEC-HMS	Physical similarity (clustering by flexible-beta)	Model parameter
Sharghi, Nourani, Soleimani, and Sadikoglu (2018)	United States	27	Monthly	—	Physical similarity (clustering)	Catchment classification
Tegegne and Kim (2018)	Korea, Ethiopia	13	Daily	SWAT	Spatial proximity, physical similarity	Model parameter
Teutschbein et al. (2018)	Sweden	14	Daily	HBV	Regression-based	Hydrological signatures
Worland et al. (2018)	United States	224	Annually	—	Machine learning, spatial proximity, regression-based	Hydrological signatures
Yang, Magnusson, Rizzi, et al. (2018)	Norway	118	Daily	WASMOD	Spatial proximity, physical similarity, regression-based	Model parameter
Zhang et al. (2018)	Australia	605	Daily	SIMHYD	Regression-based	Hydrological signatures
Alipour and Kibler (2019)	United Kingdom, United States, Australia, China	6	Daily	HyMOD	SPED framework	Model parameter
Arsenault et al. (2019)	Mexico	30	Daily	GR4J, MOHYSE, HMETs	Spatial proximity, physical similarity, regression-based	Model parameter
Betterle, Schirmer, and Botter (2019)	United States	413	Daily	A physically based model	Spatial proximity, physical similarity	Model parameter
Chouaib et al. (2019)	United States	73	Monthly	SAC-SMA	Physical similarity	Hydrological signatures
Choubin et al. (2019)	Iran	40	Monthly	IHACRES, SWAT	Physical similarity	Observed streamflow
de Lavenne, Andreassian, Thirel, Ramos, and Perrin (2019)	France	1,305	Daily	GRSD	Spatial proximity	Model parameter
Ecrepont, Cudennec, Ancil, and Jaffrezic (2019)	Canada	21	Daily	—	Geomorphology-based	Hydrographs
Jayatilake and Smith (2019)	United States	438	Daily	PDM	Hydrological signature methods	Model parameter
Konrad and Rumsey (2019)	United States	1,019	Daily	—	Physical similarity, regression-based	Minimum streamflow
Kratzert et al. (2019)	United States	671	Daily	SAC-SMA, NWM	Machine learning	Observed streamflow
Kuzmin et al. (2019)	Vietnam, Cambodia	3	Daily	MLCM3, SAC- SMA, HBV	Physical similarity	Catchment classification
Lee and Kim (2019)	Korea	1	Hourly	HEC-HMS	Regional frequency analysis	Flood quantiles
Pagliero, Bouraoui, Diels, Willems, and McIntyre (2019)	Europe	246	Monthly	SWAT	Physical similarity (clustering)	Model parameter
Prieto et al. (2019)	Spain	92	Daily	PDM	Regression-based	Model parameter

(Continues)

TABLE A1 (Continued)

First author	Country	Stations/ catchments	Time scale	Hydrological model	Regionalization methods	Regionalization target
Saadi, Oudin, and Ribstein (2019)	France, United States	2,225	Hourly	GR4H	Machine learning	Model parameter
Silva, Blanco, and Pessoa (2019)	Brazil	43	Daily	—	Regression-based	FDCs
Swain and Patra (2019)	India	32	Daily	SWAT	Spatial proximity, physical similarity	Model parameter
Worland, Steinschneider, Farmer, Asquith, and Knight (2019)	United States	74	Daily	Copula theory	Spatial proximity	Observed streamflow
Zamoum and Souag-Gamane (2019)	Algeria	64	Monthly	GR2M	Physical similarity (clustering using PCA and SOM)	Model parameter
Yilmaz and Onoz (2020)	Turkey	15	Daily	DAR method	Physical similarity	Observed streamflow